



STOCKEDUP

Online Shopping (E-Commerce) System

A Project Presentation | MERN Stack Web Application

Presented by

Group Leader: Shubham Kr. Mandal

Members: Aditya Raj (Regd. No. 2501445017)

Rudra Narayan Barik (Regd. No. 2501445603)

Chanchal Patar (Regd. No. 2501445488)

Live Project: <https://stockedup.vercel.app>

Project Overview

Stockedup is a full-stack e-commerce website. Customers can browse products, add them to the cart and place orders, while an admin dashboard is used to manage the store. Built with the MERN stack.

- Two parts of the application - customer storefront and admin dashboard
- Product catalogue with 168 products in 6 categories, with search and filters
- Login/signup using JWT tokens; passwords stored in encrypted (hashed) form
- Cart, coupon-based discounts and order placement with e-mail confirmation
- Responsive design, works on mobile as well as desktop
- Hosted on Vercel, database on MongoDB Atlas (cloud)

Problem Statement

Why we built this project -

- 1) Small/local shops have a limited reach - they can only sell to nearby customers
- 2) Using existing marketplaces means paying high commission and platform fees
- 3) Customers waste time on slow search and unclear product information
- 4) Order and inventory handling is done manually, which is slow and error prone
- 5) Online stores are common targets of hacking - passwords and data need protection
- 6) There was no simple and affordable e-commerce solution for smaller stores

Objectives

Main Objectives

- Build an easy-to-use online store for browsing, searching and buying products
- Provide separate access for customers and admins using secure login
- Allow cart management, coupon discounts and order placement with confirmation
- Give admins a dashboard to manage products, categories and orders

Technical Objectives

- REST API with clear separation of client and server code
- Secure password hashing and JWT-based sessions
- Responsive UI that works on all screen sizes
- Cloud deployment on Vercel with MongoDB Atlas

Final deliverables: working web application, source code package and project report

Technology Stack

We used the MERN stack - JavaScript is used on both frontend and backend.

Layer	Technology	Used for
Frontend	React 18 + React Router + Vite	User interface, pages, routing
Backend	Node.js + Express	REST API, logic, authentication
Database	MongoDB (Atlas) + Mongoose	Stores users, categories, products, orders
Security	JWT + bcrypt	Login tokens and password hashing
Hosting / Tools	Vercel, Git, NPM	Deployment, version control, packages

System Architecture

CLIENT React SPA - user interface, pages, API calls

JSON over HTTPS

APPLICATION SERVER Node.js + Express - REST API, auth, business logic

Mongoose queries

DATABASE MongoDB Atlas - users, categories, products, orders

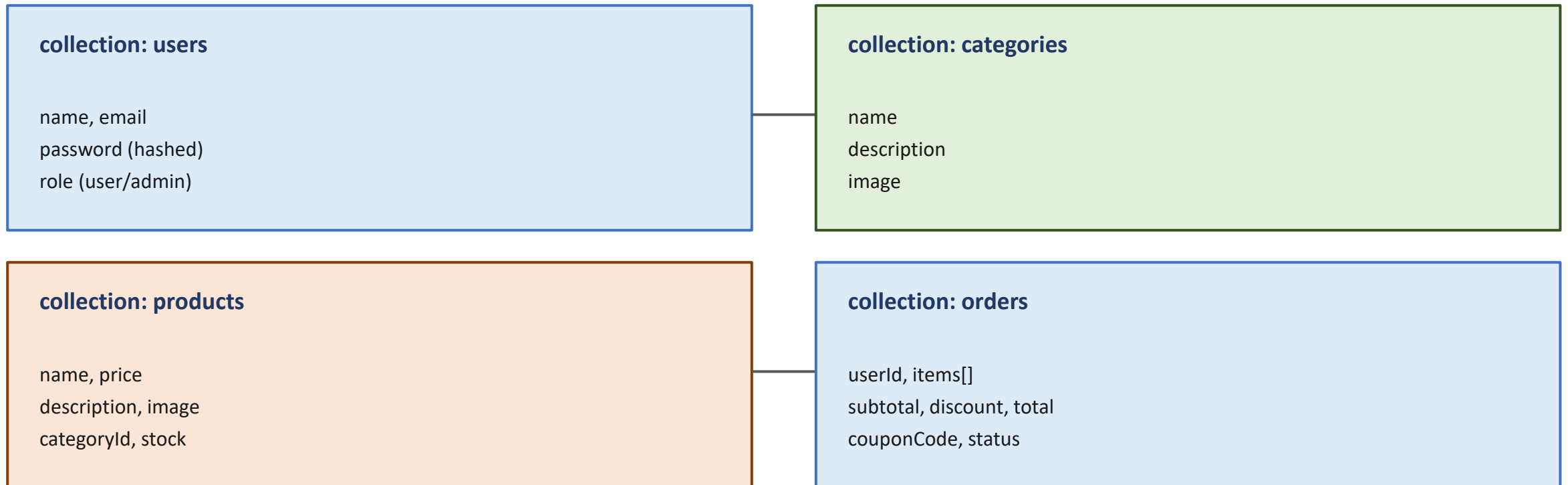
Important API calls Products & search -> GET /api/products, /api/categories

Login & register -> POST /api/auth/login, /api/auth/register

Place order -> POST /api/orders (can apply coupon code)

Database Design

MongoDB is a document database. This project uses 4 collections -

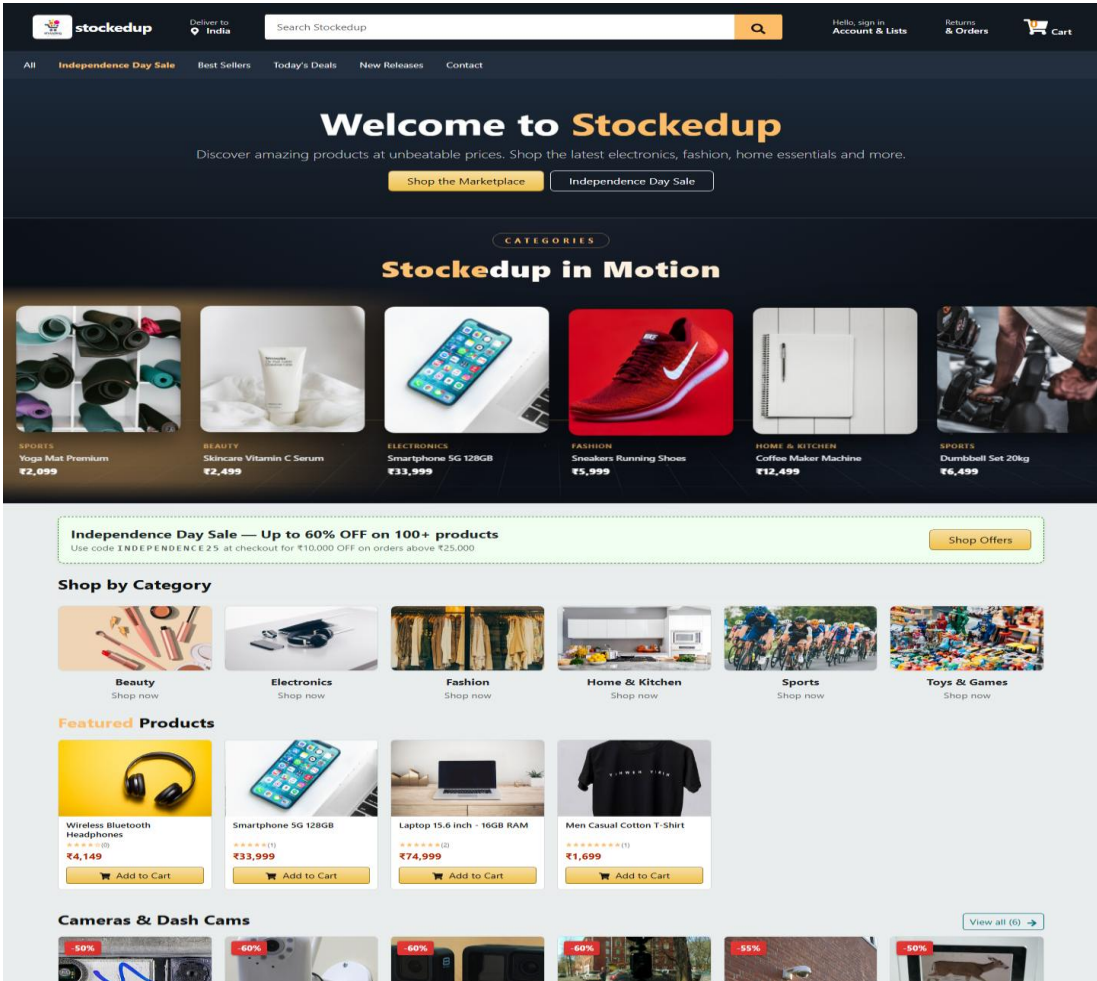


Relations: products.categoryId -> categories orders.userId -> users orders.items[] -> products

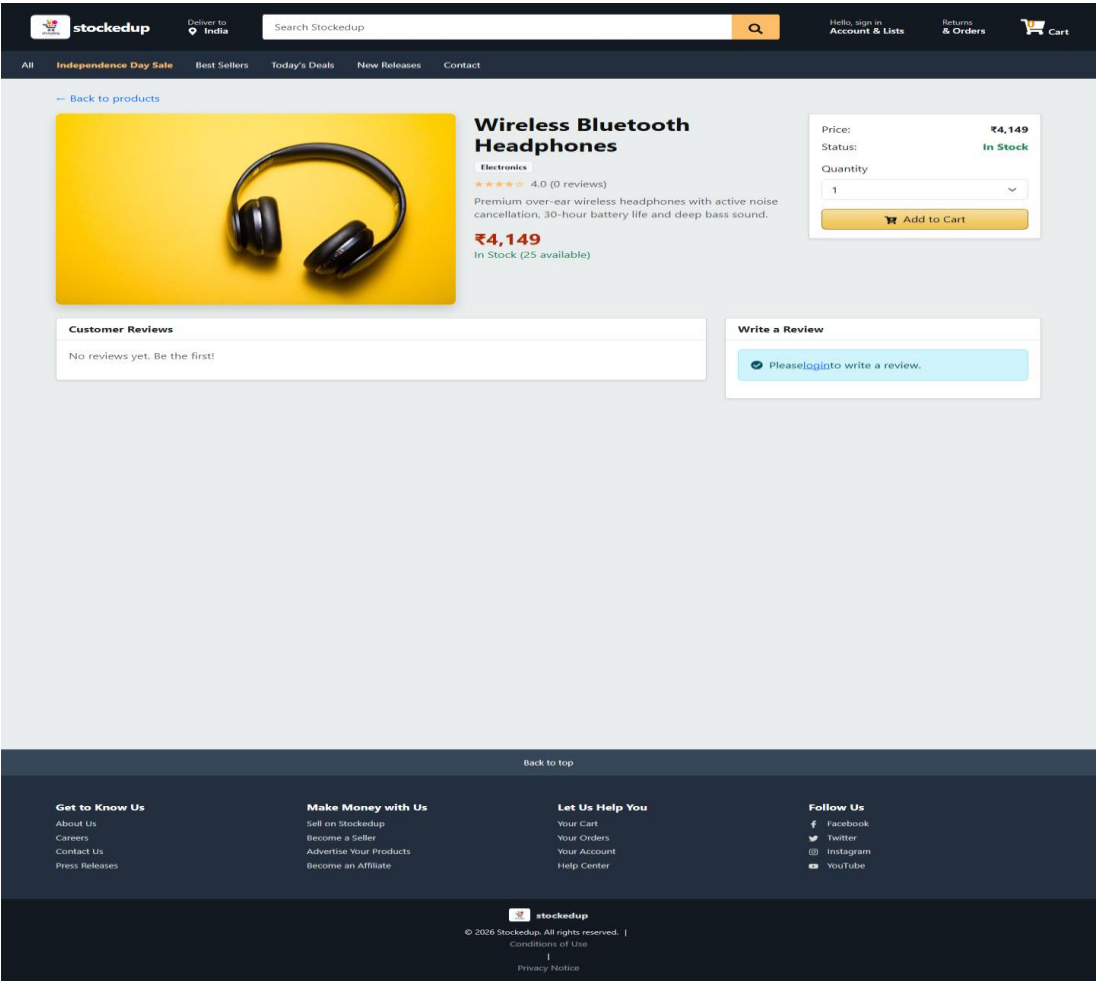
Main Modules

Module	Description
Authentication	Register / login for customers and admin, JWT based security
Product catalogue	Browse, search and filter 168 products in 6 categories
Cart & checkout	Add/remove items, apply coupon (INDEPENDENCE25), place order
Orders	Order placement with e-mail confirmation, admin can update status
Admin dashboard	Manage products and categories, view all orders
Other pages	Contact page, about, offers, 404 error page

Screenshots - 1

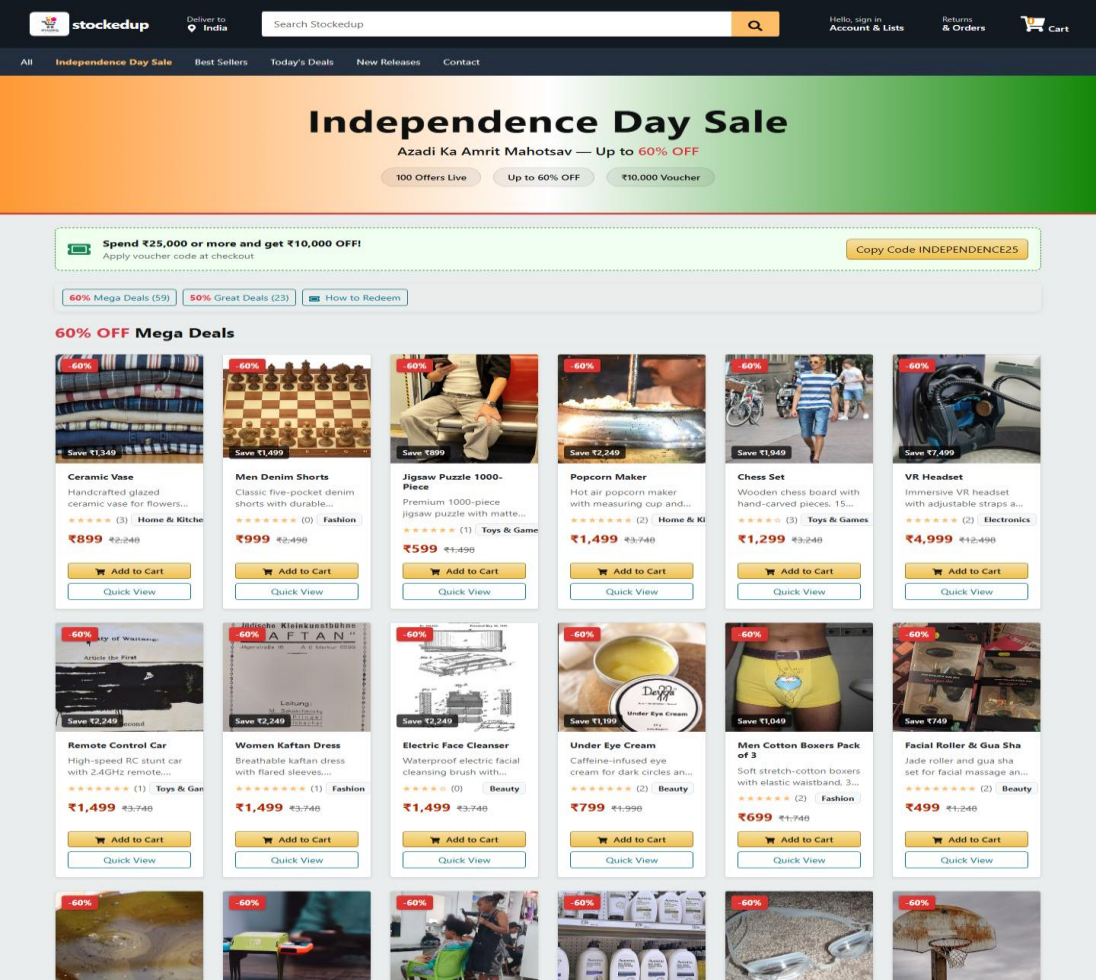


Home page - storefront

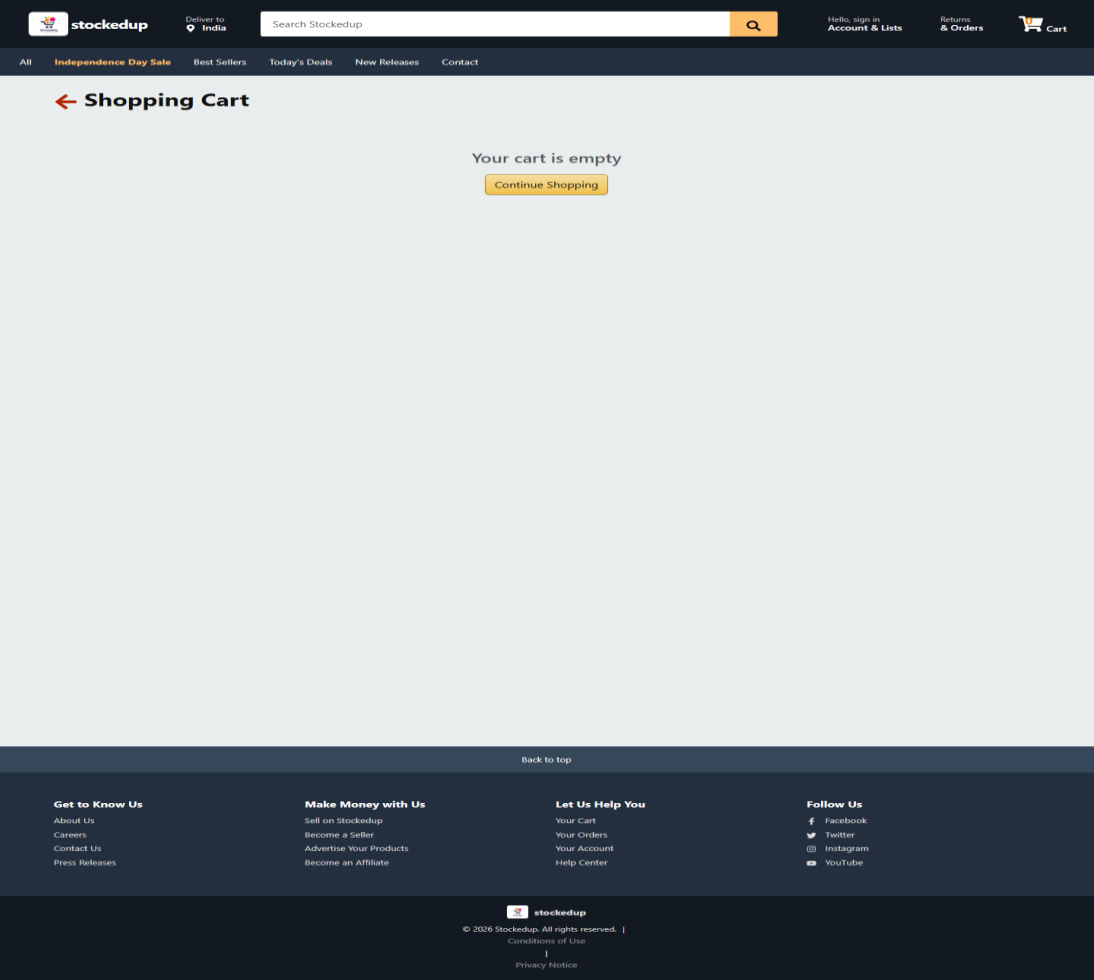


Product detail page

Screenshots - 2



Offers page with coupon INDEPENDENCE25



Cart page

Testing

- API testing - 16 test cases for auth, products, categories, cart, coupons and orders - all passed
- Functional testing - 10 user journeys tested on storefront and admin dashboard - all passed
- Responsive testing - checked the layout on mobile, tablet and desktop sizes
- Security testing - wrong passwords, fake tokens and unauthorised admin access are rejected
- Fixed bugs - carousel fallback when images fail, zoom issue on iPhone (small input font), slow page loading due to large images

Overall: all test cases passed before final deployment.

Conclusion & Future Scope

Conclusion

The project was completed successfully. Stockedup works as a full e-commerce website - customers can shop and place orders, and admins can manage the store. The application is live on Vercel and was tested thoroughly before submission.

Future Scope

- Online payment gateway (UPI / cards) for real transactions
- Order tracking, wishlist, reviews and ratings for products
- Better inventory management, stock alerts and sales reports
- Cloud storage for product images
- Installable mobile app support (PWA)



THANK YOU

Any questions?

Group Leader: Shubham Kr. Mandal

Members: Aditya Raj (Regd. No. 2501445017)

Rudra Narayan Barik (Regd. No. 2501445603)

Chanchal Patar (Regd. No. 2501445488)

Department of CSE | Gandhi Engineering College (GEC Autonomous), Bhubaneswar • <https://stockedup.vercel.app>